

Non-parametric regression

1 Exercise 1

The dataset `engine`, in the package `gamair` collects the capacity (variable `size`) and wear (variable `wear`) of 19 engines.

- a) Plot the data (`wear` as a function of `size`) and propose a parametric model to describe their relationship. What do you observe ?
- b) As the linear model fails to capture the complex structure of the data, we go for a non-parametric approach. We are going to construct the nonparametric fit "manually" based on the piecewise linear (tent) bases. Please refer to course slides, p.273-274. We provide some useful functions in the file `HW09Functions.R`. You can import them in your R environment using the command `source("HW09Functions.R")`.
 - 1) Using the provided function `tf.X()`, build the design matrix X , based on a grid of evenly spaced knots.
 - 2) Estimate the parameter β by least squares using the function `lm()` (without the intercept).
 - 3) Try different number of knots, and plot the predictions. Explain the role of the knots, and the limitations of this basis.

2 Exercise 2

We usually need the model's function $f(\cdot)$ to be smooth. Thus, the piecewise linear model might be too coarse. To this aim, a polynomial basis would be more suitable than the "tent" one. For computational stability, we would rather choose orthogonal basis functions.

- a) Simulate the data using the following code:

```
set.seed(1)
x <- sort(runif(40)*10)^0.5
y <- sort(runif(40))^0.1
```

- b) Plot y against x .
- c) Fit the 5th and 10th order parametric polynomial models. First, compute manually the OLS estimator with the model matrix defined in the course (slides 271-272). Then, use `lm()` with the same model matrix in the formula. Comment on the findings.
- d) This time, use the function `poly()` in order to define an orthogonal polynomial basis, and the corresponding model matrix. Then, repeat point **c**) and explain the changes.
- e) Plot the predictions of the two polynomial fits (using `predict()`) on a sufficiently fine grid along with the data, what do you observe ?
- f) The function `cub_X()` (available in `HW09Functions.R`) allows to construct the model matrix corresponding to the cubic splines. Use it with evenly spaced knots to fit the cubic splines smoother with `lm()`.
- g) Overlay the cubic splines predictions on your previous graph. Comment the fit of the different approaches.

3 Exercise 3

The `temperature.txt` data file features the average minimum January temperature in degrees Fahrenheit according to the latitude (in degrees north of the equator) and longitude (in degrees west of the prime meridian) of 56 U.S. cities. For each year from 1931 to 1960, the daily minimum temperatures in January were added together and divided by 31; These averages were computed for the 30 years ¹.

- a) Analyse the data with the `gam()` function in package `mgcv`.
- b) Try different number of knots, smoothness parameter, and possible spline basis. On which argument does the fit mainly depend?

¹Data source: Peixoto, J.L. (1990) A property of well-formulated polynomial regression models. *American Statistician*, 44, 26-30.